Character Emotion Arcs

Sentiment and emotion arcs by character: how the emotional register around each character rises and falls across a narrative.

Contents

What an emotion arc is

An emotion arc traces the emotional register of a text against narrative position — the shape of the story, rather than its content. The idea is Kurt Vonnegut's: he argued in a rejected anthropology thesis that stories have simple shapes that can be drawn on graph paper ("Man in Hole", "Boy Meets Girl", "Cinderella"). Reagan et al. (2016) tested the claim computationally on 1,327 Project Gutenberg texts and found that emotional arcs are dominated by six basic shapes.

The NLP Suite's Character Emotion Arcs takes that one step further, and asks the question a reader of a novel would ask: not "what shape is this story?" but "what shape is this story FOR THIS CHARACTER?" — whose fortunes rise, whose fall, who is surrounded by fear while someone else is surrounded by joy, and where the arcs cross.

What the tool actually does

Three steps, in this order:

1. Stanza's named-entity recogniser reads the corpus sentence by sentence and picks out every PERSON it finds. A sentence naming two characters belongs to both of them; a sentence naming none is filed under _NARRATOR/UNATTRIBUTED_, so nothing is thrown away.
2. Every sentence is scored on the eight NRC emotions with the NRCLex lexicon — anger, anticipation, disgust, fear, joy, sadness, surprise and trust. The eight scores are normalised to sum to 1, so what is plotted is the SHARE of a sentence's emotion words carried by each emotion, not a raw count.
3. Each character's sentences are put in narrative order and smoothed with a centred rolling mean over 5 sentences, which is what turns a scatter of sentence scores into a readable arc.

English only. The tool uses an English NER model and an English lexicon, and it says so and stops if the corpus language is set to anything else.

How characters are identified — and merged

Names are normalised per document. The first form of a name encountered becomes the canonical one, and any later name that CONTAINS it, or is contained BY it, is folded into it. So "Harry", "Harry Potter" and "Mr. Potter" collapse sensibly — and so, less sensibly, do "Anna" and "Anna Maria". Check the Character column of the output CSV before you trust a character's arc: it is the one place where you can see exactly who has been merged with whom.

Where to find it

Two routes, both running the same code:

- **Sentiment analysis GUI.** This dropdown lists it under "Character-level"; it also has its own checkbox — "Do sentiments fluctuate across actors and/or locations? (Sentiment scores by NER values)".
- **Corpus Profiler.** It is one of the analyses the Profiler runs. There it is much faster: the Profiler has already produced a Stanza NER table for the corpus, and the arcs are derived from that table instead of parsing the corpus a second time. A full re-parse of a corpus the size of the Harry Potter novels takes about an hour and a half; deriving from the existing table takes minutes.

What you get

Everything is written to a character_emotion_arcs subfolder of your output directory:

Output	What it holds
character_emotion_arcs…csv	One row per (document, sentence, character), with the sentence itself and its eight emotion scores. This is the data behind every chart, and the file to open when a chart surprises you.
emotion_arc_<Character>.png	One chart per character: all eight emotions plotted against narrative position, in the standard NRC colours.
dominant_emotion_timeline_<Character>.png	A ribbon showing which single emotion is strongest at each point — the same data as the arc chart, read as a sequence of moods rather than eight curves.
character_comparison_<emotion>.png	The same emotion for all the leading characters on one chart, so their arcs can be compared. Written for joy, anger, fear and sadness whenever there are at least two characters.
character_emotion_summary_….csv	One row per character: how many sentences they appear in, their average score on each of the eight emotions, and their dominant emotion overall.

Which characters get charts

The leading five, by number of sentences they are named in, among those named in at least five sentences. The narrator/unattributed rows are excluded from that ranking — they are almost always the largest group, and they are not a character. Those thresholds (five characters, five sentences, a five-sentence smoothing window) are the tool's defaults and are not currently exposed in the GUI.

Reading the charts

- **The x axis is narrative position:** the character's sentences in order — the 1st, 2nd, 3rd sentence they are named in. It is NOT time, and it is not the sentence number in the text: a character absent for fifty pages leaves no gap on their own chart.
- **The y axis is intensity:** the share of that sentence's emotion words carrying that emotion, smoothed over five sentences. Values sum to 1 across the eight emotions before smoothing, so a rise in one emotion is always a fall in others. Read the SHAPE, not the height.
- **Flat stretches are usually silence, not calm:** a sentence with no emotion words at all scores zero on everything and pulls the smoothed line down. Long stretches of plain narration flatten an arc.

What it does not tell you

The single most important caveat, stated plainly:

The emotion belongs to the SENTENCE the character is named in, not to the character. A character mentioned in a frightening sentence is not necessarily afraid — they may be the one causing the fear, or merely present. What the arc measures is the emotional weather AROUND a character, which is a real and interesting thing, but it is not that character's feelings.

Beyond that:

- The NRC lexicon counts words. It does not handle negation ("not afraid" counts as fear), irony, or figurative language.
- Only named characters are found. A character referred to throughout as "the old woman" or "he" is invisible to NER — see the TIPS on coreference resolution for what would be needed to fix that.
- The name-merging rule described above can over-merge, and NER can miss a name entirely in a sentence where it appears in an unusual form.
- Run one document at a time when you want a narrative arc. Sentence IDs restart in each document, so a corpus of several files has its sentences interleaved along the x axis and the arc becomes an average of several narratives rather than the shape of any one of them.

Getting more out of it

- Compare a character's arc with the whole-corpus sentiment series from any of the other sentiment algorithms: divergence between a character and their story is often exactly the point.
- Use the comparison charts to find the crossing points — where one character's joy overtakes another's — and then open the CSV at those sentence numbers and read what actually happens there. The chart tells you where to look; the text tells you why.
- The summary CSV is the fastest way into a large corpus: sort by dominant emotion and see which characters are carried by fear, and which by trust.

The arcs as an interactive file

Beside the per-character PNGs the tool writes emotion_arcs_<corpus>.html: the same arcs, with a character and an emotion to choose. Open it in any browser; it needs no internet and no libraries, so it travels with the rest of the output.

The PNG is still written, and it is the one to paste into a paper. The HTML is the one to explore.

Why the static chart is hard to read

Eight emotions are plotted over every appearance a character has. For Hermione that is about 5,000 appearances, so the chart carries roughly 40,000 points in twelve inches. The eight lines cross constantly and the result is a band of colour in which no single arc can be followed. Nothing is wrong with the numbers, and binning alone does not save it:

eight lines still cross constantly however few points each one has. So the PNG is now drawn as SMALL MULTIPLES - one small panel per emotion, four rows by two, each showing that emotion in colour with the other seven behind it in grey and every panel on the same scale. A single line against a common scale is readable, which one pair of axes carrying eight of them is not.

What the interactive file does differently

Each character's appearances are grouped into 120 equal-count stretches and each stretch draws its MEAN, so a line turns 120 times instead of 5,000 and its shape survives being drawn. The grouping does not move the level: a count-weighted mean of the stretches equals the character's overall mean, the same figure reported in character_emotion_summary_<corpus>.csv.

Equal-COUNT stretches, not equal-width: equal widths would be empty wherever the character is absent for a stretch of the narrative, and a gap in a line chart reads as calm rather than as absence.

Choosing one emotion brings it to the front and drops the other seven to faint grey, so they stay as context instead of competing. The strongest stretch of the chosen emotion is ringed. Moving across the chart reads out all eight values at that point, ordered strongest first, with the number of sentences behind them. The legend switches an emotion off entirely, and "scale to the chosen emotion" re-scales the axis to a quiet emotion that would otherwise sit flat along the bottom.

Reading the numbers with care

Two cautions, both of which matter more than the chart makes them look.

FIRST: an emotion score belongs to the SENTENCE the character appears in, not to the character. The NRC lexicon scores the words of the whole sentence, so a calm character standing in a frightening room scores fear. This is a measure of the emotional colour of the prose around a character, which is a real and interesting thing, but it is not the character's own feeling and should not be reported as one.

SECOND: characters in the same book tend to come out looking alike. Across the seven Harry Potter books, Harry, Ron and Hermione all rank the eight emotions in exactly the same order - anticipation, trust, fear, sadness, anger, joy, surprise, disgust - and their averages differ in the third decimal place. That ordering is largely the book's vocabulary rather than a difference between the characters. Compare the SHAPE of an arc over the narrative, and compare characters against each other at the same moment; treat a small gap in overall averages as saying little.

A single sentence can score 1.00 on one emotion. Scores are normalised over the eight emotions, so a sentence matching exactly one emotion word gives that emotion the whole of the sentence. Those spikes are why the averages, and the 120-stretch means, are the figures to read rather than individual sentences.

References

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See also the NLP Suite TIPS on Sentiment Analysis, and The world of emotions and sentiments.